Chemical, Biological, Radiological and Nuclear (CBRN) Defense Research Institute
Ministry of National Defense, Seoul, Republic of Korea
2 November 2026

Dear Editors,

We submit *Descriptor Similarity Does Not Predict Array Response: Chemical Warfare Agent Simulants Tested Against Seventeen Authentic Agents* for consideration as an Article.

What the paper is about, and why we bring it here. Chemical warfare agent detection is almost always developed and qualified on simulants, because the agents themselves are available to few laboratories. Every such qualification rests on one assumption: that a compound chosen to resemble an agent on molecular descriptors will resemble it in what the detector measures. We tested that assumption directly, against colorimetric response measured on seventeen authentic agents spanning nerve, blood, choking, vesicant and A-series compounds. This journal sets the standard for whether a detection claim has been demonstrated against the analyte it names. The question we ask is one step earlier and bears on every simulant-qualified device in the literature: whether the surrogate analyte was the right one to begin with.

Three counts state the result. Across the seventeen agents, exactly one designated simulant has ever been measured beside them, and that single co-measurement sits in one run of one array. Twelve exact Mantel tests and six Fisher combinations of them leave no agreement between descriptor distance and response distance below $q = 0.05$. On the thirty-five-compound panel, no agent of eleven has a valid match to an incumbent simulant, at a Benjamini–Hochberg adjusted value of 0.6173. The array itself is not in question: it classifies these agents at a cross-validated accuracy of 0.964 on the seven-agent run and 1.000 on the Novichok run. What fails is the transfer from descriptor space to response space.

The specific case will interest sensor developers. Dimethyl methylphosphonate is the standard G-agent simulant and the benchmark compound for decontamination efficiency. Measured beside the agents in the ultraviolet channel, its response pattern correlates at $+0.76$ with tabun and $+0.65$ with sarin but at -0.49 with soman, -0.72 with cyclosarin and -0.70 with VX. Its similarity to sarin also falls with concentration, from $+0.77$ at 500 μM to $+0.36$ at 10 μM, which is the end at which a field detector operates. A surrogate that resembles the agent only at high dose is of limited use to the measurement it stands in for.

What this paper is not, stated plainly. It reports no new sensor, no demonstration in a complex matrix, and no correlation against an established analytical method, and we recognise that your guidelines ask for those of a sensing paper. This is not a sensing paper. It is a measurement-science audit of the surrogate on which sensing papers are qualified, and the data it audits could not be produced any other way: the agents cannot be put into a complex environmental matrix outside a designated laboratory, and the comparison it makes is between two descriptions of the same compounds rather than between a sensor and a reference method. We would rather you weigh that against your scope now than have it surface in review.

Data provenance. The colorimetric response data are reanalysed from three earlier studies by this group, in *Journal of Hazardous Materials*, *Discover Sensors* and *Sensors and Actuators B*, all cited in the manuscript. Those papers reported classification performance for three separate agent sets. None of them compared response to molecular descriptors, none placed the three sets in one analysis, and no figure, table or passage is reproduced here. The descriptor calculations, the permutation tests and every result in this manuscript are new.

What we propose instead of a replacement compound. We propose none, because selecting one on descriptor similarity would repeat the error the paper is about. What the manuscript specifies is the measurement that would settle the question: a bridging run in which candidate simulants and the agents they

stand in for are exposed on the same plate, in the same optical channels, at the same concentrations, within one session. That run yields a response-space distance between a simulant and its agent measured under identical conditions, which no amount of descriptor work can supply. It needs no new chemistry and no new instrument.

Reporting and availability. All agents were produced and handled at an Organisation for the Prohibition of Chemical Weapons designated laboratory under Chemical Weapons Convention procedures. Analysis code, descriptor tables, parsed regulatory values with their sources and every figure script are deposited in a public repository cited in the manuscript, and all permutation tests are exhaustive rather than sampled, so every number reported can be reproduced from the deposit. The manuscript has not been published elsewhere and is not under consideration at another journal. The authors declare no competing financial interest.

Yours sincerely,

Ku Kang, on behalf of the authors
Corresponding author: Yoon Jeong Jang, yjjangchem@korea.kr